

GROW Computing · paper route evidence map

Teacher guide · AQA Unit Award Scheme 71638, Programming with Scratch (unit 6), Level One

This states, outcome by outcome, how a pupil who took the PAPER route evidences the same outcome as a pupil who took the screen route.

The published evidence requirement for every outcome is a summary sheet completed by the teacher from individual observation. Pupil work supports that record; it does not replace it. So the question for each outcome below is whether a teacher can OBSERVE the pupil doing the required thing on paper.

Record the route a pupil took HERE, in the teacher guide. Never on the AQA summary sheet, which is completed identically whichever route the pupil took.

Eight lessons alone do not establish that every pupil has completed the unit. The centre's AQA summary sheet remains the official record.

Outcome 1 · Week 3 · Build the world

What the outcome asks, in the pack's own words:

AQA 71638 outcome 1, in the teacher guide's own words: "Observe each pupil creating and selecting all three sprites. Record what they did and the support used. A downloaded finished maze does not establish creation." Lesson goal, in the pack's own words: "I can create a maze, a player and a finish as three sprites." Teacher guide expected answer 1: "The three roles are Maze, Player and Finish. A maze painted only as a backdrop does not meet this outcome."

What the teacher observes on the paper route:

- Date, pupil name, and the route taken: paper route. Record the route here in this teacher guide.
- Observed the pupil making three separate cards. Record what they drew on each one.
- Observed the pupil naming each card. Record the method: written, traced over the model word, or dictated and scribed word for word.
- Observed the pupil pick out each named card on request. Four asks across three names. Record how many were right first time, and whether the pupil lifted the card or pointed to it.
- Observed the lift test. The Maze card lifted and the walls came with it. The plain sheet stayed on the table. Record the pupil's answer to "sprite or backdrop".
- Observed the pupil move a counter from their own start, through both gaps, to Finish, recording the coordinates reached. Copy the pupil's own start coordinates into this guide. Weeks 4, 5 and 6 use them.
- Observed the pupil compare the 16-unit counter and the 80-unit counter at the 60-unit gap, and give a fits / does not fit answer.
- Observed the pupil close the W03_Maze wallet, reopen it, and name all three cards again.

- Record the support used against each of the four pupil checks, exactly as the guide's observation table asks: "Date and support used".
- Do not sign any statement from a handed-in card set you did not watch being made. The pack's rule holds on paper exactly as on screen: a finished maze does not establish creation.
- Complete the AQA summary sheet as published, from this observation. Do not write the route on it. Do not annotate it. Do not reprint it. The route a pupil took is recorded in this teacher guide only.

What the pupil records:

- ☐ I created and named Maze, Player and Finish. I made all three cards myself. I wrote, traced or said each name.
- ☐ My Maze is a sprite I can select. When an adult says Maze, I pick out the Maze card. My walls come up with the card. The plain sheet stays on the table.
- ☐ Player fits through the gaps. I moved my counter through gap 1 and gap 2 and wrote down where it went. I tried the big counter too. It did not fit.
- ☐ I saved and reopened my project. I put my three cards and my sheets in my wallet. I wrote W03_Maze on the wallet. I closed it, opened it again, and named all three cards.

Named limits of the paper route for this outcome:

Outcome 1 itself is carried in full on paper, and I would not write NONE only because one named thing is not. The outcome's two assessed acts - creating all three sprites, and selecting each one - are performed by the pupil and watched by the teacher, and the published evidence requirement is a summary sheet completed by the teacher from individual observation, not a file or a screenshot. The misconception the outcome exists to catch is caught more sharply on paper than on screen: lifting the Maze card and watching the walls come away from the plain sheet is a direct physical test of sprite versus backdrop, where on screen it is an inference from the sprite pane. One thing is genuinely not carried, and it must be named rather than dressed up. The pack's fourth check, "I saved and reopened my project", means a Scratch file operation on the screen route: File > Save to your computer, then File > Load from your computer, producing W03_Maze.sb3. The paper route substitutes a named project wallet that is closed, reopened and re-identified. That evidences what the check exists to prove for Outcome 1 - that the pupil's own three sprites persist and can be named again afterwards - but it does not evidence Scratch file handling, and it produces no .sb3 file. No outcome in AQA 71638 requires file handling or a .sb3 file, so Outcome 1 is not weakened by this. Two consequences follow and should be recorded rather than hidden. First, if a school's own internal policy - not AQA's - asks for a digital artefact per pupil, the paper route will not produce one. That is a policy matter to settle with the UAS lead before teaching, not a reason to defer the outcome to a device session. Second, a pupil who begins on paper and later moves to the screen route has no W03_Maze.sb3 to open, so they build their three sprites in Scratch at that point. That is a fresh construction on a new route, not a repeat of an outcome already evidenced. Outcome 1 must not be re-assessed, and must not be held open, because of it.

Outcome 2 · Week 4 · Make controls reliable

What the outcome asks, in the pack's own words:

AQA 71638 outcome 2: Observe pupil creation of all four arrow scripts and a practical demonstration. Check any previously inherited controls by having the pupil construct the relevant script.

What the teacher observes on the paper route:

- Observed the pupil select Player: the pupil pointed to or named Player from the three sprite headers (Maze, Player, Finish) before writing anything.
- Observed the pupil place the counter on a square clear of every wall, read its x from the printed x axis and its y from the printed y axis, and write that same pair into the go to x y block. Record the pupil's actual safe start; (-180,-120) is the supplied model example only.
- Observed the pupil write each of the four key scripts on its own separate grid, choosing the motion block from the word bank. The block was not named for them and was not copied from the teacher answer page.
- Observed that right and left use change x, and up and down use change y, in the scripts the pupil wrote.
- Observed the pupil reset the counter to their own start square before each single test, move the counter one square for the key given, and record the resulting x and y. The recorded pairs agree with the script the pupil wrote.
- Observed the practical demonstration: reset, then right, up, left, down in one sequence, the counter returning to the start square, with the four rows of Trace 2 completed.
- Observed the pupil repair the printed up-arrow fault by naming the axis (change y), not by changing the sign.
- Observed the pupil name the hat block for each of the five jobs in the Which hat block starts it? table, so the event was chosen by the pupil and not only read off the printed grid.
- Observed the pupil take the sheets back out of the folder and repeat one key test from their own writing, unaided.
- Any inherited or copied work was re-checked by having the pupil construct the relevant script again on a fresh grid, unaided.
- Support used and communication method (pointed, said, drew, wrote, or exact words scribed) recorded against each row.
- Route taken: PAPER. Record this here in the teacher guide only. Do not write the route on, or alter, the AQA summary sheet.
- Record the completed outcome on the AQA summary sheet from this observation, exactly as for a screen route. No later device opportunity is required for this outcome.
- All four controls remain the target for every route; two keys are partial progress.

What the pupil records:

- ☐ I made four separate key scripts for Player. I wrote each one myself.
- ☐ My right and left scripts change x. My up and down scripts change y.
- ☐ My reset uses a start square that is clear of every wall. I wrote its x and y.
- ☐ I moved my counter for every test. I wrote the new x and y each time.
- ☐ I reset, then did right, up, left, down. My counter came back to my start.

- ☐ I found the fault on the up key card. I said the axis was wrong, not the number.
- ☐ I put my sheets in my folder. I took them out again. I tested my controls again from my own writing.
- ☐ Tick, point to, or say yes for each line. An adult can write your words for you.

Named limits of the paper route for this outcome:

One thing does not transfer, and it is the pack's fourth check: "I saved, reopened and tested my controls." Half of that check is a file operation - File then Save to your computer, a file named W04_Controls.sb3, a school folder on a school system. On paper no .sb3 exists, so that half cannot be evidenced and must not be ticked as though it were. The other half - that the work persists, is reopened and still runs - is evidenced and observed: the sheets go into the pupil's folder, come back out at 35 to 40 minutes, and the pupil repeats one key test from their own writing, unaided. AQA 71638 outcome 2 as published (observe pupil creation of all four arrow scripts and a practical demonstration) is carried whole by paper, because it asks for creation and for demonstration and never asks for a file; the evidence AQA requires is a summary sheet completed by the teacher from individual observation, and every one of those observations is available here. So the named gap is the file-handling wording inside check 4, not the outcome. The teacher must record in the teacher guide, not on the AQA summary sheet, that the route was paper and that check 4 was met as folder, reopen and retest. Two smaller things, said plainly. First, the hat block is printed at the top of each grid, so the pupil chooses the motion block and the numbers but does not produce the event block from an empty page. The Which hat block starts it? table exists to close that, and the teacher should observe the pupil name the key rather than read it off the grid; if a pupil only copies the printed hat, that part is not evidenced. Second, on paper a wall cannot physically stop the counter, so a pupil can trace a move straight through a wall and still be marked correct - but that is equally true on screen in Week 4, because collision is the Week 5 task, so it is not a weakness of the paper route.

Outcome 3 • Week 5 • Make a rule work

What the outcome asks, in the pack's own words:

AQA 71638 outcome 3: Observe the pupil constructing the collision script, then demonstrate contact at two wall positions. Record the actual consequence and any prompts.

What the teacher observes on the paper route:

- I watched this pupil write the collision script block by block in the Player grid. I did not give them a finished script to copy. I did not use the teacher model as something to copy from.
- I watched the pupil put if then inside forever, and put the return block inside if then. The nesting is shown by the indents, or by the boxes the pupil drew, or by the pupil pointing at the two spaces and telling me which block goes in each.
- I watched the pupil name Maze in the sensing block, not Finish.
- I watched the pupil demonstrate contact at TWO DIFFERENT wall positions with the counter. Position one: x -200, y -120 (the left wall). Position two: x 50, y 0 (the right inside wall). These are two different walls, not the same wall twice.

- THE ACTUAL CONSEQUENCE I OBSERVED: at each contact the pupil moved the counter back to their own start, x ____ , y ____ . I recorded what actually happened, not what should have happened.
- I watched the pupil check a clear space and show me that the return did NOT run there.
- PROMPTS: I recorded, against each pupil action, whether it was unprompted, or given a verbal prompt, or given a modelled step. I did not leave the prompt box blank.
- I recorded the pupil's access method: writing / speaking / pointing / drawing / directing an adult to move the counter / exact words scribed. Where the pupil directed an adult, the decision was the pupil's and the adult only carried it out.
- I recorded whether the pupil went back to the check row on their own, or whether I told them to.
- ROUTE TAKEN: PAPER. This pupil completed Week 5 on paper. That is recorded here, in this Teacher Guide, and nowhere else.
- I have not written on, altered or reprinted the AQA Summary sheet in any way.
- I completed the AQA Summary sheet for outcome 3 from my own individual observation of this pupil. I did not infer creation from supplied code.

What the pupil records:

- ☐ I built a repeated touching Maze check on Player. I wrote it in my Player grid: when green flag clicked, then forever, then if touching Maze? then go to my start.
- ☐ The consequence runs only when the condition is true. I showed this on my trace sheets. On my No rows the go to block did not run. On my Yes rows it did.
- ☐ I tested two wall positions and a clear space. Wall one was x -200, y -120. Wall two was x 50, y 0. My clear space was x 0, y 0 and my clear walk up from the start.
- ☐ I saved and reopened the working rule. I wrote my name and the date on my sheets. I put them in my folder. Later I found my own sheets on my own. I traced one wall test again from my own writing, and it still worked.
- ☐ I said what I think about the return place. An adult wrote what I said. I know what changed because of it.

Named limits of the paper route for this outcome:

ONE GAP, NAMED PLAINLY.

On a screen, the forever loop makes the COMPUTER repeat the check by itself, many times a second, with the pupil doing nothing. On paper, the PUPIL repeats the check. So the paper route evidences that the pupil understands repeated checking. It does not show a machine performing that repetition.

Why this matters in practice: a pupil could fill in a trace table correctly and still believe the check happens only once, because on paper they are told when to check.

Two things in this route close that gap, and both are required, not optional. First, the Compare table. The pupil traces the broken one-check example and their own forever script from the SAME counter walk, and writes the two different endings. A pupil who thinks the check happens once cannot produce two different endings and explain why.

Second, the observation itself. The teacher must watch whether the pupil goes back up to the 'if touching Maze?' row without being told, and must write 'unprompted' or 'prompted' in the prompt box. Write the word. Do not leave it blank. That single word is what separates evidenced understanding from a neatly filled table.

ALSO PLAINLY, so nobody is surprised later: the paper route produces no .sb3 file. The pack check 'I saved and reopened the working rule' is met here by the pupil writing their name and the date on their sheets, filing them, later finding their own sheets without help, and re-tracing one wall test from their own writing. That is a genuine save and reopen of the pupil's own work, and AQA outcome 3 does not require a file - the published evidence requirement is a Summary sheet completed by the teacher from individual observation. But if your centre has separately told a moderator that a digital artefact will exist for every pupil, paper will not produce one. Settle that with the UAS lead BEFORE the pupil starts this route, not after.

Outcome 4 • Week 6 • Finish and restart

What the outcome asks, in the pack's own words:

AQA 71638 outcome 4: Observe pupil construction and demonstration of the win script. Record a win and a restarted run. (Teacher guide, Individual evidence.) Audit outcomes 1-4 individually. Do not claim outcomes 5-6 from the teacher model.

What the teacher observes on the paper route:

- Construction, observed directly: the pupil chose wait until from the word bank, not forever, and wrote touching [Finish]? inside the hexagon in grid 1 by their own hand.
- Construction, observed directly: the pupil joined one say block below wait until, and left the first say block empty.
- Construction, observed directly: the go to block in grid 1 uses the same x and y as the pupil's start script in grid 7.
- Demonstration, observed directly: the pupil traced their own script in Table 2 and moved their counter square by square in Table 3, and did not say the win message until the first square where touching Finish was true.
- Demonstration, observed directly: exactly one win message per run. The pupil did not repeat it while the counter stayed on Finish.
- Demonstration, observed directly: on the green flag the pupil turned the message card face down and returned the counter to the safe start, in that order.
- Demonstration, observed directly: after the restart the pupil placed the counter on x -60 y 0, applied their own wall rule and returned the counter to x -180 y -120, then reached Finish a second time and gave one message.
- Independent execution, observed directly: a partner or adult followed the pupil's written script for Table 5 without being told what the pupil meant. Record whether the script needed any spoken clarification before it could be followed.
- Record the date and the support used against each of the four pupil checks, in the Individual evidence table of this teacher guide.
- Record here that this pupil took the PAPER ROUTE for Week 6, and that no W06_Win.sb3 exists. Week 7 must reprint this pupil's grid 1 page instead of opening a file.

- Record completed outcomes on the AQA summary sheet. Write nothing on the AQA summary sheet about which route the pupil took. The route is recorded in this teacher guide only.

- Do not infer creation from the printed listings on the challenge page. Only grid 1, written in the lesson, evidences creation.

What the pupil records:

- ☐ I created a wait until condition that checks touching Finish.
- ☐ My win message appears once per green-flag run.
- ☐ The flag clears the message and returns Player to the start.
- ☐ The controls and wall rule still work, and I can win again after restart.
- ☐ I wrote the win script in grid 1 myself.
- ☐ My safe start is x _____ y _____. My win message is _____.
- ☐ I moved my counter and I filled in Table 3.
- ☐ Someone else followed my written script in Table 5. I watched them.
- ☐ My idea, and what it changed: _____.
- ☐ Tick, point to, or say each line you agree with. An adult can write your words.

Named limits of the paper route for this outcome:

Not NONE. Two things are named plainly, so a moderator can see them rather than find them. (1) NO .sb3 FILE EXISTS ON THIS ROUTE. The pack's Exit steps 'Save as W06_Win.sb3', 'Load the saved file' and 'File > Save to your computer' cannot be done with paper. Those steps are file handling. They are not outcome 4. Outcome 4 asks for construction and demonstration of the win script and for a win and a restarted run to be recorded. All of that is done and observed on these pages, so the outcome itself stands whole. What is genuinely lost is the pack's continuity artefact: Week 7 expects to open the pupil's Week 6 file. On this route the pupil's dated grid 1 page in their folder is that artefact instead, and the teacher guide must say so, or Week 7 will ask for a file that was never made. (2) ON SCREEN THE COMPUTER EXECUTES THE SCRIPT. ON PAPER A PERSON DOES. A computer cannot be persuaded and does not guess. A pupil who writes an unclear script may trace it as they meant it rather than as it is written, and a kind adult may quietly repair it while following it. This route answers that with Table 5: someone else follows the pupil's written script exactly, doing only what is written, while the pupil watches and does not prompt. That is a real control and a standard one, and it catches the ordinary mistakes this week teaches, such as a say block placed above wait until. But it is a person applying it, not a machine. A teacher signing outcome 4 on this route is stating that they watched the pupil write the wait until condition, and watched a faithful trace of that written script produce one message and a clean restart. That is the honest scope of the claim. It is the same claim a teacher makes when they watch a screen, because the published evidence requirement is a summary sheet completed by the teacher from individual observation either way.

Outcome 5 · Week 7 · Week 7 Add one challenge — paper completion route (no device)

What the outcome asks, in the pack's own words:

AQA 71638 Programming with Scratch (unit 6), Level One. Outcome 5: added a feature that changes gameplay. The pack's teacher guide states the evidence: "Observe the pupil creating the feature and explaining what the player must now do. Record a before-and-after test. Decoration alone does not meet the outcome." Evidence for the outcome is a summary sheet completed by the teacher from individual observation. The centre's AQA summary sheet remains the official record and is completed by staff. The statements below are a local scaffold only. The route the pupil took is recorded in the teacher guide, never on the AQA summary sheet.

What the teacher observes on the paper route:

- Watch the pupil write Grid 1 and Grid 2. The pupil must place both glide blocks inside forever, and must place the contact check on Player and not on Hazard. Ask which sprite the go to block belongs to. The answer must be Player.
- Watch the pupil name the Hazard. Naming and drawing must be the pupil's own, not copied from a printed model. Record which parts the pupil produced today.
- Ask the discriminating question: what must the player now do that they did not do before? A pupil who answers only "it looks different" has described decoration, and outcome 5 is not met. A pupil who says they must watch the Hazard, wait, or go round the side has met it.
- Record the before-and-after test from the before and after table. The same setup, two different results, is the required record.
- POSE AT LEAST ONE UNSEEN SETUP. The four shared setups have their expected results printed in this guide, so a pupil could copy them. An unseen setup shows the pupil is applying the rule. Use one of these three. U1: green flag, Player counter on (0, 110), advance the clock 4 steps. U2: green flag, Player counter on (30, -110), advance the clock 8 steps. U3: green flag, Player counter on (0, -110), do not move the clock at all.
- Accept pointing at the counter, saying the answer, drawing the move, or writing it. Any of the four is a full response.
- Record the date and the support used against each pupil statement.
- Record in the teacher guide that this pupil took the paper route. Do not write the route on the AQA summary sheet and do not alter that sheet.
- There is no W07_Challenge.sb3 on this route. Initial the pupil's dated Week 7 pages in the folder. Week 8's paper route opens those pages.

What the pupil records:

- ☐ I created a new sprite and named it myself. On paper: I drew the Hazard on the design sheet, I wrote the name Hazard at the top of its own block grid, and I cut out and placed its counter.
- ☐ My feature moves or responds on every run, not only once. On paper: I wrote forever with both glide blocks inside it, and I ran the clock strip twice all the way round to show the journey comes back to (0, -100) and goes again.
- ☐ Contact with my feature has a consequence the player can see. On paper: I traced a contact setup, I moved the Player counter back to (-180, -120), and I said out loud why it moved.
- ☐ The four controls, the wall rule and the win still work after my change. On paper: I wrote out all four arrow scripts, the wall rule and the win response, and I traced one arrow run, one wall square and one win.

☐ Extra, from the design sheet: I can say what the player must now do that they did not do before.

☐ Extra, from the compare table: I chose my glide setting from a test result I wrote down, not from a feeling.

Named limits of the paper route for this outcome:

Five things are genuinely not evidenced on paper, and staff should not claim them. FIRST, no Scratch sprite is created. Pupil check 1 says "I created a new sprite and named it myself". On paper the pupil designs, draws, names and places a Hazard, and the teacher can observe every part of that. What is not observed is the pupil using the Scratch sprite pane. If a centre reads outcome 5 as requiring the sprite to be made inside Scratch, the paper route does not supply that one act, and the pupil should repeat only that step when a device is next available. SECOND, and this is the real difference between the two routes, no program is ever executed by a computer. On paper the pupil is the interpreter. The teacher observes the pupil applying its own rule correctly and consistently, including moving the Player counter back to (-180, -120) on contact. What is not observed is a built program producing that consequence by itself. This is exactly why the unseen setup above is not optional: without it, a pupil who copies the printed expected results has demonstrated nothing, and the observation would not hold. With it, the observation does hold, because the pupil has to apply the rule to a setup they have not seen an answer for. THIRD, no file exists. The Exit step on the screen route is save, reload, run one check. That has no paper equivalent and must not be ticked. The paper equivalent is dated pages filed and initialled, and it evidences filing, not saving. FOURTH, continuous motion is not seen. The clock strip shows the same start, the same turning point and the same return in half-second steps, and running it twice round shows the journey repeats. The pupil has not watched a glide. This does not affect outcome 5, which asks what the player must now do, but nobody should claim the pupil watched the Hazard move. FIFTH, the paper play rule, five small squares of Player to one clock step of Hazard, is a fair-play convention chosen so both move ten small squares a second. In Scratch a held arrow key repeats faster than that. The paper game is therefore harder than the screen game at the same settings, and the paper timings are not a measurement of Scratch. One thing does hold exactly, and is worth stating: the paper touching rule and the screen touching rule agree at every grid crossing. Same crossing or a neighbouring crossing, corner to corner included, is touching. Two crossings apart is not. This was checked against the model's own arithmetic at every position on the lattice, so paper and screen give the same answer to every setup in these tables.

Outcome 6 • Week 8 • Week 8 • Test and improve • Paper completion route (no device)

What the outcome asks, in the pack's own words:

AQA 71638 Programming with Scratch (unit 6), Level One, outcome 6: "Tested the game." The pack's own teaching wording is: "Observe the pupil testing the game, recording expected and actual results, and retesting after a repair." The pack also states, twice, that "AQA summary sheets remain the official evidence" and "Record completed outcomes on the AQA summary sheet. Do not infer creation from supplied code." The five-check record, the written expected and actual notes and the improvement are named in the pack as local teaching scaffolds: "outcome 6 asks for testing and does not require finding a fault", and "An all-pass result is valid." Week 8 also carries the whole-unit review: outcome 6 is claimed here and outcomes 1 to 5

are reviewed individually using

Planning/GROW_Computing_Final_Evidence_Check.docx. "A partly finished game gets a recorded next step, not an assumed completion."

What the teacher observes on the paper route:

- Row 1 of the teacher guide Individual evidence table - "I ran all five checks using the named setups." On paper: the pupil put the counter cross on the named coordinates before each check, and named or pointed to the setup each time.
 - Row 2 - "I recorded expected and actual results, not only 'it works'." On paper: the pupil gave the expected result before the trace and the traced result after. Both are on the five check record, in the pupil's words or in a scribe's hand marked as scribed.
 - Row 3 - "I recorded a fault and repair, or an all-pass result and useful improvement." On paper: for a repair, the pupil wrote or dictated the old line and the new line on the block grid. For an all-pass, the pupil wrote or said all pass and named one improvement.
 - Row 4 - "I retested the same steps and the earlier checks still pass." On paper: the pupil traced the repaired script again from the same named setup, then traced check 3 and check 4 again to show the repair broke nothing.
 - Fidelity note for the adult. Read each block aloud exactly as written. Never say the direction. The pupil moves the counter. If the pupil moves the counter the way the block was meant rather than the way it is written, stop and read the block again. The trace is the evidence. A trace steered by the adult is not evidence.
 - Route note. Record here, in this teacher guide, that the pupil took the paper route and why. Do not write the route on the AQA summary sheet. Do not alter any awarding-body document.
 - Save note. On this route no W08_Final.sb3 exists. Record the dated and signed pages and their folder location instead. Do not enter a filename that was never made.
 - Authorship note. The Test Copy is supplied material. Record what the pupil traced, found and repaired on it. Do not infer creation from supplied code, and do not infer it from a printed script the pupil only copied out.
 - Whole-unit review. Complete
- Planning/GROW_Computing_Final_Evidence_Check.docx for outcomes 1 to 6, using the pupil's dated paper pages as the file or observation reference. Leave an outcome pending where the demonstration is missing, and write the next step.

What the pupil records:

- ☐ I ran all five checks on my own game pages, using the named setup each time.
- ☐ I said or wrote what I expected before I traced it, and the traced result after.
- ☐ I traced each block exactly as it is written, not as I meant it to be.
- ☐ I recorded a fault and a repair, or an all-pass result and one useful improvement.
- ☐ I traced the same five checks again after the change, and the checks that passed before still pass.
- ☐ I asked a tester to trace my script from a named start, and I wrote their exact words.

Named limits of the paper route for this outcome:

Outcome 6 is met on paper. Three limits are real, and are named here rather than hidden. FIRST, a paper trace samples time; Scratch runs it continuously. Check 4 and the Challenge edge case "contact during a glide" are traced at half-second steps, so the Hazard is seen at y -100, -50, 0, 50, 100 and back. A contact that would happen between two of those instants is not seen on paper. The check still evidences expected against actual and a retest, but it evidences a sampled journey, not a continuous one. Use that wording in the teacher guide. SECOND, paper can only find a fault that is already visible in the written script. On a device the machine is an independent authority and will surface faults nobody wrote down. On paper the pupil executes only what is on the page, so a mismatch between what the pupil wrote and what the pupil meant is findable, and a mismatch between what the pupil wrote and what Scratch would actually do with it is not. This is exactly why the Test Copy is supplied with three faults already written down: it gives the paper route something real to find and repair. THIRD, the Exit save-and-reopen loop has no paper equivalent, because no .sb3 file exists on this route. The substitute is: the pupil dates and signs the pages, puts them in the folder, the folder is closed, and the pupil reopens it and traces one named check from the pages alone. That evidences that the record is complete enough to work from later. It does not evidence a saved digital file, and the teacher guide should say so in plain words rather than record a filename. ONE THING THAT IS NOT A GAP. If the pupil's own game passes all five checks, the repair and the retest are demonstrated on the Test Copy rather than on their own game. That is identical on the screen route, where an all-pass pupil also repairs nothing of their own. The pack states that outcome 6 asks for testing and does not require finding a fault, and that an all-pass result is valid. ONE RISK THAT MUST BE MANAGED, NOT WRITTEN AWAY. The trace is only evidence if it is faithful. A pupil who moves the counter the way the block was meant has restated an intention, not tested a script. Observation must therefore be closer on paper than on screen: the adult reads the block and never the direction, and watches the counter move. If the adult supplies the direction, the outcome is not evidenced and must be left pending with a recorded next step.

Weeks 1 and 2

Week 1 • Get a result

No AQA outcome is claimed this week. Week 1 is foundation practice. The pack's own words: "These lessons build foundations for AQA 71638 Programming with Scratch (unit 6), Level One. Neither Week 1 nor Week 2 completes the full unit." What this week evidences is the pack's own core goal: "run a program, change its movement value, retest, then save and reopen the changed project." Pupil-facing: "I can run a program, change one value and save my work." The record completed is the pack's own five-row Individual observation record: "Open the correct project." / "Predict and compare a movement result." / "Change the value and explain its effect." / "Run twice from the known start." / "Save, locate and reopen the changed project."

For the outcome this week evidences, NONE. No AQA outcome is claimed in Week 1, and every row of the pack's five-row Individual observation record is directly observable on paper as set out above. Predicting, tracing block by block, running twice from a returned start, naming, filing, finding and re-running are all things a teacher watches the pupil do.

One thing is genuinely not evidenced, and it is honest to name it because it is not an outcome and must not be mistaken for one. The Scheme of Work sets a Week 1-2 diagnostic: "Confirm pointer/keyboard access, file handling and a workable prompt level in Weeks 1-2." Paper confirms the prompt level and confirms who is making the decisions. Paper does not confirm that a pupil can use Scratch's File > Save to your computer, find a downloaded .sb3 in a school folder, or operate a mouse or keyboard. That is a device-operation check, not a unit outcome, and AQA 71638 does not assess it.

It only matters for a pupil who later moves from the paper route to the screen route. A pupil who stays on the paper route for all eight weeks never needs it. Record the diagnostic as "not applicable, paper route" rather than as a gap in achievement. If a move to the screen route is planned, book the file-handling check before Week 3, because Week 3 is where the first AQA outcome is claimed.

Week 2 • Control a character

Week 2 assessment destination, in the pack's own words: Foundation practice. "These lessons build foundations for AQA 71638 Programming with Scratch (unit 6), Level One. Neither Week 1 nor Week 2 completes the full unit." No AQA outcome is claimed this week. The work here evidences the pack's own core completion statement for Week 2: "The pupil creates and tests BOTH right and left scripts. One working script is an early milestone and leaves a recorded next step. All four controls are checked in the main maze in Week 4." The shared goal is: "I can use key events to move a sprite and fix a direction mistake." This feeds outcome 2, "Create keyboard arrow-control scripts", which is claimed in Week 4 and not before.

One of the pack's five checks differs in kind between the two routes, and the teacher record must say so. Check 5 is "Save, locate and reopen the controls." On paper the pupil names the pack W02_MyControls, dates it, files it, finds it again, rereads both grids and retraces one control to the same finish. That evidences what the check is for: the pupil keeps their own work, finds it again without help, and confirms their

scripts are still there and still behave the same. It does not evidence the Scratch operation itself - File, Save to your computer; an .sb3 file on disk; File, Load from your computer. A pupil on the paper route has not done those things, and a teacher must not record that they have. Two reasons this does not leave the week incomplete. First, Week 2 claims no AQA outcome; the pack states plainly that neither Week 1 nor Week 2 completes the unit, so no assessed evidence is lost. Second, none of the six 71638 outcomes is a save-and-reopen outcome. The outcome Week 2 feeds is outcome 2, "Create keyboard arrow-control scripts", and creating and testing both control scripts is exactly what the paper route evidences. So the honest record for check 5 on this route reads "named, filed, found and retraced the paper controls", and that is a complete record of what was observed. A record that instead read "saved and reopened a .sb3" would be the false mapping this route exists to avoid, because nobody reading it later could tell it was untrue. One much smaller difference, named for completeness and costing nothing: a pupil on paper never meets key auto-repeat, so the misconception that a held key moves the sprite several times never arises for them. The press-and-release rule is still taught, as "one press is one square".

The one thing to decide locally

Week 7, outcome 5. On the paper route a pupil designs, draws, names and places the Hazard, and writes its blocks, and every part of that is observed. What is not observed is the pupil using the Scratch sprite pane.

The outcome as published asks the pupil to add something that makes the game more complex, and to explain what the player must now do. It does not name a tool. If your centre reads outcome 5 as requiring the sprite to be made inside Scratch, the paper route supplies every other part and that one act should be repeated when a device is next available. Record that decision here, once, for the cohort — not per pupil, and not on the AQA form.